

Paper II - Urology (MCh Urology, Summer 2015)

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II-1. Describe medical management of BPH and role of various surgical modalities in its management.

Type: LAQ (25 marks) | ~1500 words

Introduction / Definition

- **BPH** = histological diagnosis (stromal + epithelial hyperplasia of transition zone); **BPE** = enlargement; **BPO** = obstruction from BPE; presents as **LUTS** (storage, voiding, post-micturition).
- Goals of therapy: relieve bother, improve QoL, prevent progression (AUR, surgery, renal damage).
- Sources: Campbell: BPH – evaluation, medical and surgical management chapters; **AUA guideline 2021, amended 2023**; **EAU Non-neurogenic Male LUTS guideline**.

Pathophysiology relevant to therapy

- **Static component** – prostate bulk (androgen/DHT dependent) → target of 5-ARIs, surgery.
- **Dynamic component** – α 1A-adrenergic smooth-muscle tone in prostate/bladder neck → target of α -blockers.
- **Bladder response** – detrusor overactivity/underactivity → antimuscarinics, β 3-agonists.

Evaluation (brief)

- History, **IPSS** (7 questions, score 0–35: mild 0–7, moderate 8–19, severe 20–35 + 1 QoL question 0–6), frequency-volume chart (3 days, for nocturia/polyuria).
- DRE, urinalysis, **PSA** (if life expectancy >10 y or if it changes management; also surrogate of volume).
- Optional: serum creatinine (if renal impairment suspected), uroflowmetry (Q_{max} <10 mL/s suggests BOO), PVR.
- **Before surgery** (AUA): measure prostate size and shape (TRUS/TAUS/cystoscopy/CT/MRI) – choice of modality depends on it; PVR, Q_{max} ; pressure-flow study if diagnosis uncertain (e.g. Q_{max} >10, neurological disease, young men, large PVR).

Medical management

1. Watchful waiting / behavioural (WW)

- For mild or non-bothersome symptoms: education, fluid timing, reduce caffeine/alcohol, double voiding, urethral milking, bladder retraining, review drugs (diuretics, anticholinergics, decongestants).

2. α 1-adrenergic blockers (first line for moderate-severe bothersome LUTS; act within days)

Drug	Dose	Notes
Terazosin	1→10 mg OD (titrate)	Non-selective; orthostatic hypotension; bedtime dosing
Doxazosin	1→8 mg OD (GITS 4–8 mg)	Non-selective; titrate
Alfuzosin	10 mg OD (ER)	"Clinically uroselective" – least ejaculatory dysfunction
Tamsulosin	0.4 mg OD	α 1A/ α 1D selective; ejaculatory dysfunction
Silodosin	8 mg OD (4 mg in renal impairment)	Most α 1A selective; highest anejaculation

- Improve IPSS ~30–40% and Qmax modestly; **do not reduce prostate volume or AUR/surgery risk** in long term.
- Side effects: dizziness, postural hypotension, asthenia, nasal congestion, ejaculatory dysfunction.
- **IFIS (Intraoperative Floppy Iris Syndrome)** – billowing iris, progressive miosis, iris prolapse during cataract surgery; strongest with tamsulosin; **advise men planning cataract surgery to defer starting α -blocker** (stopping beforehand does not reliably prevent it; inform ophthalmologist).

3. 5 α -reductase inhibitors (5-ARIs)

- **Finasteride 5 mg OD** (type 2 isoenzyme); **Dutasteride 0.5 mg OD** (types 1 and 2).
- Indicated with **enlarged prostate** (AUA: palpable enlargement, PSA >1.5 ng/mL or volume >30 mL; EAU: >40 mL).
- Reduce volume by ~**20–25%** over 6–12 months; **halve PSA** (multiply by 2 for interpretation after 6 months); reduce AUR and BPH surgery risk by ~50%.
- Also used for BPH-related haematuria (finasteride).
- Side effects: decreased libido, ED, ejaculatory disorder, gynaecomastia. PCPT and REDUCE showed increased detection of high-grade cancer → FDA label warning (2011).

4. Combination α -blocker + 5-ARI

- **MTOPS (McConnell, NEJM 2003)**: doxazosin + finasteride vs either vs placebo; combination reduced overall clinical progression risk by **66%** vs placebo (doxazosin 39%, finasteride 34%); only finasteride/combination reduced AUR and invasive therapy.
- **CombAT (Roehrborn, 2010, 4-yr)**: dutasteride + tamsulosin in men with volume $\geq$ 30 mL, PSA 1.5–10; combination superior for symptoms; reduced AUR/BPH surgery by ~**66% vs tamsulosin** (not significantly different from dutasteride alone).
- Indicated in men with **enlarged prostate at risk of progression**; consider stopping α -blocker after 6–12 months.

5. Drugs for storage symptoms

- **Antimuscarinics** (solifenacin 5–10 mg, tolterodine 4 mg ER, fesoterodine, oxybutynin) – alone or add-on to α -blocker for persistent OAB; caution if **PVR >250–300 mL** (retention risk); cognitive side-effects in elderly (anticholinergic burden).
- **β 3-agonists**: mirabegron 25–50 mg OD, vibegron 75 mg OD – add-on to α -blocker; no cognitive burden; monitor BP (mirabegron).

6. PDE5 inhibitor

- **Tadalafil 5 mg OD** – improves LUTS ($\pm$ ED), not Qmax significantly; contraindicated with nitrates; combination with α -blocker not recommended by AUA for LUTS (no added benefit, hypotension).

7. Nocturnal polyuria

- **Desmopressin** (low-dose; oral/sublingual/nasal) for nocturia due to nocturnal polyuria; check **serum sodium** at baseline, within first week and periodically (hyponatraemia risk higher in elderly).

8. Phytotherapy

- Saw palmetto (*Serenoa repens*), *Pygeum africanum*, beta-sitosterol – AUA: **not recommended** (no proven benefit; CAMUS/STEP trials of saw palmetto negative). EAU: hexane-extracted *Serenoa* may be offered with caution.

Male LUTS → evaluation (IPSS, DRE, U/A, PSA, FVC)
 Mild/non-bothersome → Watchful waiting + lifestyle
 Bothersome:
 Voiding predominant → α -blocker ($\pm$ 5-ARI if prostate >30–40 mL)
 Storage persists → add antimuscarinic / β 3-agonist (PVR check)
 Nocturnal polyuria → desmopressin (Na monitoring)
 With ED → tadalafil 5 mg
 Failure / complications / patient choice → Surgery (size-based)

Surgical management

Indications (AUA 2023 / EAU)

- Absolute: **refractory urinary retention** (failed TWOC), **recurrent UTI**, **recurrent gross haematuria** refractory to 5-ARI, **bladder stones**, **renal insufficiency/upper tract dilatation** due to BPO, large bladder diverticula (relative).
- Relative: moderate–severe bothersome LUTS failing or declining medical therapy.

Modalities (sizes per AUA 2021/2023 guideline)

Modality	Prostate size	Advantages	Disadvantages
TURP – monopolar (M-TURP) / bipolar (B-TURP)	30–80 mL	Reference standard; durable	Bleeding, retrograde ejaculation (~65–75%), TUR syndrome (M-TURP), stricture, BNC
TUIP	≤30 mL (no median lobe)	Quick, less retrograde ejaculation	Higher retreatment rate
TUVP (vaporization)	30–80 mL	Haemostatic	Irritative symptoms, no tissue (limited)
PVP – GreenLight 180W XPS	30–80 mL	Anticoagulated/high-risk patients, day care	No histology; dysuria
HoLEP / ThuLEP / bipolar enucleation	Size-independent	Lowest retreatment, least bleeding, short stay, tissue for histology	Learning curve; transient stress incontinence; retrograde ejaculation (high)
Simple prostatectomy – open / laparoscopic / robotic	>80–100 mL	Complete adenoma removal; deal with stones/diverticulum	Morbidity, transfusion, longer stay
Aquablation (robotic waterjet)	30–80 mL ; may consider 80–150 mL	Image-guided, heat-free, better ejaculatory preservation	Post-op bleeding (needs haemostasis), GA, cost
Rezūm (WVTT)	30–80 mL	Office/day care, ejaculation preserved, treats median lobe	Delayed relief, catheter 3–7 days

Modality	Prostate size	Advantages	Disadvantages
PUL - UroLift	30-80 mL, no obstructing median lobe	Local anaesthesia, sexual function preserved	Lower efficacy, retreatment higher, implants complicate later surgery
TIUD - iTIND	25-75 mL, no obstructing median lobe	Temporary device (5-7 days), no tissue left	Modest efficacy; limited long-term data
PAE (interventional radiology)	Typically larger glands (AUA: may be offered; EAU: in selected men after counselling)	No GA, preserves sexual function	Less effective than TURP; radiation, non-target embolisation; needs IR expertise

TURP points

- M-TURP uses 1.5% glycine (hypo-osmolar) → **TUR syndrome** (dilutional hyponatraemia, fluid overload, glycine toxicity: confusion, bradycardia, hypertension, visual disturbance); risk ↑ with resection >60-90 min, large gland, capsular perforation, high irrigation pressure. Treat: stop surgery, diuretics, hypertonic (3%) saline for severe hyponatraemia, slow correction.
- **B-TURP** uses saline → virtually eliminates TUR syndrome; allows larger glands; efficacy similar to M-TURP.

Landmark trials

- **GOLIATH** (PVP 180W XPS vs TURP; 2 yr): PVP **non-inferior** to TURP for IPSS and Qmax; shorter catheter/stay.
- **WATER** (Aquablation vs TURP, 30-80 mL; Gilling 2018): non-inferior IPSS improvement, significantly less anejaculation; **WATER II**: single-arm, **80-150 mL**, durable results, acceptable safety.
- **L.I.F.T. study** (Roehrborn, 2013): PUL vs sham, 206 men – significant IPSS improvement, sexual function preserved; 5-yr durable with ~14% retreatment.
- **Rezūm II** RCT vs sham: sustained IPSS improvement to 5 years, no de-novo ED.

Ejaculation-preserving options: PUL, Rezūm, Aquablation, iTIND, PAE; ejaculation-sparing TURP/enucleation (preserving verumontanum/para-collicular tissue).

Complications of surgery (general)

- Early: bleeding/clot retention, TUR syndrome, UTI/sepsis, capsular perforation, failed TWOC.
- Late: retrograde ejaculation, ED (rare), incontinence (stress after enucleation – usually transient), urethral stricture, bladder neck contracture, residual/recurrent adenoma.

Follow-up

- Medical therapy: IPSS at 4-12 weeks (α-blocker, antimuscarinic), 3-6 months (5-ARI); then yearly; PSA adjustment on 5-ARI.
- After surgery: IPSS, Qmax, PVR at 4-12 weeks; histology review; retreatment surveillance.

Recent advances

- **Robotic waterjet (Aquablation)** expanding to >150 mL; **Optilume BPH** (paclitaxel-coated balloon dilation, PINNACLE trial) FDA-approved 2023.
- **Vibegron** approved for OAB in men with BPH on therapy (2024).
- **Early apical release/en-bloc HoLEP**, thulium fibre laser enucleation, **MOSES** holmium technology.
- Office-based MISTs (Rezūm, PUL, iTIND) for ejaculation-preserving outpatient treatment.

Diagram to draw: (1) McNeal zonal anatomy showing transition zone enlargement; (2) Size-based surgical algorithm (≤30 TUIP → 30-80 TURP/PVP/MIST → >80 enucleation/simple

prostatectomy).

Key points to remember: - IPSS 0–35: 0–7 mild, 8–19 moderate, 20–35 severe. - 5-ARI: ~25% volume ↓, PSA halved, AUR/surgery ↓ ~50%; MTOPS combination ↓ progression 66%. - AUA sizes: TUIP ≤30; TURP/PVP/PUL/Rezūm/Aquablation 30–80; Aquablation 80–150 considered; HoLEP/ThuLEP size-independent; simple prostatectomy >80–100 mL. - IFIS with tamsulosin – defer α-blocker before cataract surgery. - B-TURP (saline) eliminates TUR syndrome; GOLIATH: PVP non-inferior to TURP.

II-2. Discuss metabolic evaluation in urinary calculus disease.

Type: LAQ (25 marks) | ~1500 words

Introduction

- Urolithiasis is a systemic/metabolic disease; recurrence ~50% at 5–10 years without prevention.
- Purpose of metabolic evaluation: identify correctable risk factors → targeted dietary + pharmacological prevention.
- Sources: Campbell: Evaluation and medical management of urinary lithiasis chapter; **AUA Medical Management of Kidney Stones guideline (2014, amended 2019)**; **EAU Urolithiasis guideline (2024/25)**.

Physicochemical basis

- Stone formation requires **supersaturation** → nucleation → growth → aggregation → retention.
- Urine zones: undersaturated → metastable (inhibitors prevent crystallisation) → unstable/supersaturated (spontaneous nucleation).
- Promoters: Ca, oxalate, uric acid, low volume, abnormal pH; **Inhibitors**: citrate, magnesium, pyrophosphate, Tamm-Horsfall protein, osteopontin, glycosaminoglycans.
- **Relative supersaturation (RSS)** – computed (EQUIL2 / Litholink profiles) for calcium oxalate, calcium phosphate and uric acid from 24-h urine; used to guide and monitor therapy (target RSS <1 / reduction from baseline).

Who needs metabolic evaluation

- **Basic evaluation - every stone former** (AUA, EAU).
- **Comprehensive (24-h urine) evaluation - high-risk / interested first-time formers** (AUA: offer to high-risk and interested first-time stone formers and all recurrent).

High-risk stone formers (EAU)

- General: early onset (esp. children/teenagers), family history, recurrent stones, short interval recurrence, solitary kidney (functional/anatomical), bilateral/large stone burden, residual fragments.
- Stone type: **brushite ($\text{CaHPO}_4 \cdot 2\text{H}_2\text{O}$)**, uric acid/urate, **infection stones**, cystine, 2,8-dihydroxyadenine, xanthine.
- Diseases: **primary hyperparathyroidism**, metabolic syndrome, nephrocalcinosis, PKD, **GI disease** (Crohn's, ileal resection, jejunio-ileal bypass, bariatric surgery, malabsorption), sarcoidosis, spinal cord injury/neurogenic bladder.
- Genetic: cystinuria, primary hyperoxaluria, **RTA type 1**, Dent disease, Bartter syndrome, familial hypomagnesaemia-hypercalciuria-nephrocalcinosis.
- Drugs: topiramate, acetazolamide, indinavir/atazanavir, triamterene, sulfadiazine, high-dose vitamin C/D, calcium supplements.
- Anatomy: medullary sponge kidney, PUJO, calyceal diverticulum, horseshoe kidney, ureteric stricture, urinary diversion.
- Occupation/special groups: pilots, military, solitary kidney donors.

Basic metabolic evaluation (all patients)

- **History**: age at onset, number of episodes, family history, diet (fluid, salt, animal protein, oxalate-rich foods, supplements), medications, bowel disease/surgery, gout, bone disease, UTIs.
- **Imaging**: NCCT (burden, density – uric acid typically <500 HU), nephrocalcinosis.

- **Serum:** creatinine, **calcium** (ionized/albumin-corrected), **uric acid**, sodium, potassium, chloride, **bicarbonate** (low → RTA, diarrhoea), phosphate; **intact PTH** if serum calcium high-normal/raised or hypercalciuria (detect primary hyperparathyroidism – Ca high, PTH high/inappropriately normal); 25-OH vitamin D as needed.
- **Urine:** dipstick/microscopy – **pH** (persistently >6.5–7 suggests RTA/infection; <5.5 favours uric acid), crystals (hexagonal = cystine; coffin-lid = struvite), leucocytes, culture (urease organisms – Proteus, Klebsiella, Providencia).
- **Stone analysis** – mandatory at first episode and if recurrence on therapy/early recurrence: **infrared spectroscopy or X-ray diffraction** (preferred); wet chemistry obsolete. Polarisation microscopy acceptable adjunct.
- **Cystine screening: sodium cyanide-nitroprusside test** (qualitative; positive >75 mg/L) – confirm with quantitative 24-h cystine.

Comprehensive evaluation – 24-h urine

- **Two 24-h collections** (AUA: one or two; EAU recommends two), on usual diet, ≥2–4 weeks after stone event/intervention, stone-free and infection-free.
- Check adequacy using **urinary creatinine** (men ~20–25 mg/kg/d; women ~15–20 mg/kg/d).
- Parameters: volume, calcium, oxalate, citrate, uric acid, sodium, potassium, magnesium, phosphate, sulphate, ammonium, pH, creatinine, ± cystine; urea nitrogen (protein intake); supersaturation.

Reference / target values (adult, per day)

Parameter	Desirable / normal	Abnormality
Volume	> 2.5 L (target output)	Low urine volume <2 L – commonest finding
Calcium	<250 mg (women), <300 mg (men), or < 4 mg/kg	Hypercalciuria
Oxalate	< 40–45 mg	Hyperoxaluria
Citrate	> 320 mg (>450 F / >350 M used by some labs)	Hypocitraturia
Uric acid	<750 mg (women), < 800 mg (men)	Hyperuricosuria
Sodium	<100 mmol/d (≈2.3 g Na intake)	High salt → ↑ Ca excretion
pH	5.8–6.2	<5.5 uric acid; >6.5–7 CaP/RTA/infection
Magnesium	>70 mg	Hypomagnesiuria (GI)
Sulphate	Marker of animal-protein intake	↑ = high animal protein (acid load)
Cystine	Not normally detectable/very low	Cystinuria (homozygous typically > 400 mg/d)

Interpretation and diagnostic categories

1. Hypercalciuria

- **Absorptive** (↑ gut Ca absorption; types I–III in Pak classification – type III renal phosphate leak), **renal leak** (↑ PTH secondary, normal serum Ca), **resorptive** (primary hyperparathyroidism – ↑ serum Ca, ↑ PTH).
- Pak's fast/calcium-load test now rarely used; classification has limited therapeutic impact (thiazide works for all idiopathic hypercalciuria).
- Others: sarcoidosis, vitamin D excess, immobilisation, RTA 1, malignancy, glucocorticoids.

2. Hyperoxaluria

- **Dietary** (spinach, nuts, tea, vitamin C) – mild (45–60 mg/d).

- **Enteric** – fat malabsorption (Crohn's, ileal resection, bariatric RYGB): unbound fatty acids saponify calcium → free oxalate absorbed in colon; also hypocitraturia, low volume, hypomagnesiuria, acidic urine.
- **Primary hyperoxaluria** (PH1 – AGXT/AGT deficiency, liver; PH2 – GRHPR; PH3 – HOGA1): usually markedly elevated (often >75–100 mg/d), childhood nephrocalcinosis, ESKD/oxalosis; confirm by genetic testing; urinary glycolate (PH1).

3. Hypocitraturia

- Idiopathic, **RTA type 1**, chronic diarrhoea, hypokalaemia, high animal protein, thiazide-induced hypokalaemia, acetazolamide/topiramate, UTI.

4. Hyperuricosuria – high purine diet, gout, myeloproliferative disease, uricosuric drugs; promotes **calcium oxalate** stones (heterogeneous nucleation).

- **Uric acid stones**: main determinant is **low urine pH (<5.5)** (metabolic syndrome, diabetes, gout, diarrhoea).

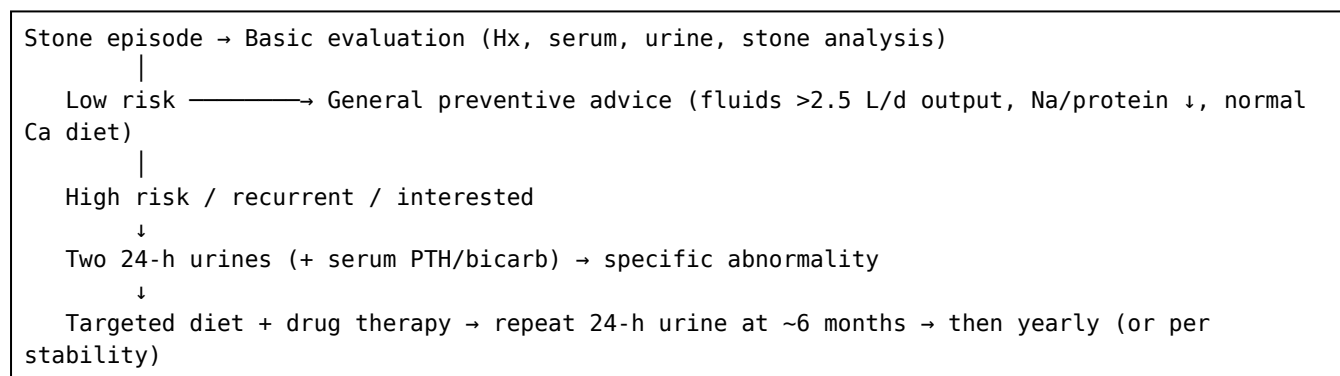
5. Distal RTA (type 1) – hyperchloraemic normal anion-gap metabolic acidosis, urine pH >5.5–6, hypocitraturia, hypercalciuria, CaP stones/nephrocalcinosis.

- Incomplete dRTA: normal serum bicarbonate → **ammonium chloride loading test** (0.1 g/kg orally; urine pH fails to fall <5.5) or **furosemide-fludrocortisone test**.

6. Cystinuria – AR defect (SLC3A1/SLC7A9) of dibasic amino acids (COLA – cystine, ornithine, lysine, arginine); cystine solubility ~250 mg/L at physiological pH → aim urine cystine <250 mg/L.

7. Infection stones – urease-producing organisms, urine pH >7, struvite/carbonate apatite; check for anatomic/neurogenic factors.

8. Hypomagnesiuria, low volume, high sodium/protein – dietary risk factors.



Treatment linked to metabolic abnormality

Abnormality	Diet	Drug
Low volume	Fluids to achieve >2.5 L output	–
Hypercalciuria (idiopathic)	Sodium <2.3 g/d, normal Ca (1,000–1,200 mg/d), ↓ animal protein	Thiazide (HCTZ 25–50 mg, chlorthalidone 25 mg, indapamide 2.5 mg) + K citrate to prevent hypokalaemia
Primary hyperparathyroidism	–	Parathyroidectomy
Hypocitraturia	↑ fruits/vegetables	Potassium citrate (30–60 mEq/d)
Hyperoxaluria – dietary	↓ oxalate, Ca with meals	–
Enteric hyperoxaluria	Low fat/oxalate, Ca supplements with meals	Calcium citrate, K citrate, cholestyramine

Abnormality	Diet	Drug
PH1	High fluid	Pyridoxine (B6-responsive), lumasiran (RNAi vs glycolate oxidase; approved 2020); nedosiran (RNAi vs LDH; PH1); liver ± kidney transplant
Hyperuricosuria (CaOx stones)	↓ purine	Allopurinol (100–300 mg); febuxostat an alternative where allopurinol not tolerated (not first-line for stones)
Uric acid stones	–	Potassium citrate to pH 6.0–6.5 (first line), ± allopurinol if hyperuricosuria
dRTA	–	Potassium citrate
Cystinuria	High fluid (>3–4 L), Na and protein ↓	Alkalinise to pH 7.0–7.5 (EAU: ≥7.5); tiopronin (preferred over D-penicillamine) if refractory
Infection stones	–	Complete stone clearance, culture-specific antibiotics, acetohydroxamic acid (limited by side effects)

Special groups

Children

- **All children with stones need full metabolic evaluation** (high prevalence of metabolic/genetic abnormalities); stone analysis essential.
- Spot urine ratios (calcium/creatinine, oxalate/creatinine, citrate/creatinine) when 24-h collection impractical (age-specific normals); 24-h urine expressed per kg or per 1.73 m².
- Screen for cystinuria, PH (genetic testing), RTA, Dent disease, infections/anatomical anomalies.

Pregnancy

- Imaging: ultrasound first, MRU (no gadolinium) second line, low-dose CT reserved.
- Defer comprehensive 24-h urine evaluation until **after delivery/lactation** (physiological hypercalciuria and hypercitraturia of pregnancy make results misleading); avoid thiazides, allopurinol, tiopronin in pregnancy.

Follow-up

- **Repeat 24-h urine within 6 months** of starting therapy to assess response (AUA), then annually (or as per stability); periodic serum chemistry for drug toxicity (K with thiazide, LFT/uric acid with allopurinol, CBC/proteinuria with tiopronin).
- Periodic imaging (US/low-dose CT) to detect new stone growth; repeat stone analysis if no response to therapy.

Recent advances

- **RNAi therapy for PH1**: lumasiran, nedosiran – marked reduction in urinary oxalate.
- Investigational: **stiripentol** (primary hyperoxaluria); oxalate-degrading bacteria/enzymes (Oxalobacter formigenes, reloxalase) for enteric hyperoxaluria.
- **Genetic panels** for monogenic stone disease in children/early-onset.
- Micro-CT, Raman spectroscopy and AI-based stone composition prediction from DECT.
- **NOSTONE trial (NEJM 2023)**: hydrochlorothiazide did not reduce recurrence vs placebo – questions routine thiazide use.

Diagram to draw: (1) Supersaturation zones (undersaturated-metastable-unstable); (2) Algorithm of basic vs comprehensive evaluation.

Key points to remember: - Two 24-h urines on usual diet, ≥ 4 weeks after event; check creatinine for adequacy. - Targets: volume > 2.5 L, Ca $< 250/300$ mg, oxalate < 40 mg, citrate > 320 mg, uric acid $< 750/800$ mg. - dRTA: urine pH > 5.5 despite acidosis; NH_4Cl load fails to lower pH < 5.5 . - Cystinuria: keep cystine < 250 mg/L, pH 7.0–7.5, tiopronin. - Repeat 24-h urine within 6 months; lumasiran for PH1.

II-3a. Management of priapism

Type: SAQ (10 marks) | ~550 words

Definition

- Full or partial erection persisting **>4 hours** beyond or unrelated to sexual stimulation (AUA/SMSNA guideline 2021).

Classification

Feature	Ischaemic (low-flow)	Non-ischaemic (high-flow)	Stuttering
Mechanism	Veno-occlusive, compartment syndrome	Unregulated arterial inflow (cavernous artery-sinusoidal fistula)	Recurrent self-limiting ischaemic episodes
Cause	ICI drugs, sickle cell disease, antipsychotics/trazodone, haematological malignancy, idiopathic	Perineal/penile trauma (straddle injury)	SCD commonest
Rigidity / pain	Fully rigid, painful	Partially rigid, painless	Painful episodes
Emergency	Yes	No	Prevent progression

Diagnosis

- History (duration, drugs, trauma, SCD), examination (glans soft in ischaemic).
- Corporal blood gas:**

	pO ₂ (mmHg)	pCO ₂ (mmHg)	pH
Ischaemic	<30	>60	<7.25
Normal arterial	>90	<40	7.40
Normal mixed venous	40	50	7.35

- Colour Doppler US:** absent cavernous flow (ischaemic); high flow/fistula ± pseudoaneurysm (non-ischaemic).
- CBC, haemoglobin electrophoresis, toxicology.

Management of ischaemic priapism

Ischaemic priapism

- Dorsal penile block → aspiration (16–18G, ±saline irrigation)
- Intracavernosal phenylephrine 100–500 µg/mL, 1 mL every 3–5 min, up to ~1 h (monitor BP/pulse; lower conc. in cardiovascular disease)
- Failure → Distal shunt → Proximal shunt
- Duration >36 h / failed shunts → consider immediate penile prosthesis

- Distal (cavernoglanular) shunts:** percutaneous – **Winter** (Tru-cut biopsy needle), **Ebbehoj** (scalpel through glans), **T-shunt** (Brant; scalpel turned 90°); open – **Al-Ghorab** (excision of tunica tips); **corporal tunnelling/snake manoeuvre** (dilator passed through Al-Ghorab or T-shunt).
- Proximal shunts:** **Quackels** (cavernoso-spongiosal), **Grayhack** (cavernoso-saphenous), Barry (deep dorsal vein).
- Oral systemic therapy alone is not recommended for acute ischaemic priapism.

- Prolonged ischaemia (>24–48 h) → corporal fibrosis, ED in most; **early prosthesis** (AUA: may offer for priapism >36 h or failed shunts) is easier and preserves length vs delayed implant in fibrotic corpora (higher infection/erosion in acute setting – counsel).
- **SCD**: intracavernosal therapy first; **concurrent** haematology care (hydration, O2, exchange transfusion) – but do not delay urological intervention.

Non-ischaemic priapism

- Not an emergency; **observation** first (many resolve spontaneously).
- Persistent/bothersome → **selective super-selective embolisation** (absorbable gelfoam/autologous clot preferred – lower ED; coils permanent).
- Surgical ligation of fistula rarely (when embolisation fails).

Stuttering priapism

- Treat each acute episode as ischaemic.
- Prevention: **daily PDE5i** (paradoxical, in non-erect state), pseudoephedrine, self-injection phenylephrine; hormonal – GnRH agonists/antiandrogens (adults, not for those desiring fertility); in SCD – hydroxyurea, transfusion protocols.

Recent advances

- SCD-targeted agents (e.g. crizanlizumab) being studied for priapism prevention; penile prosthesis early in refractory cases.

Diagram to draw: Penile cross-section showing distal (glans-corpora) vs proximal (corpora-spongiosum) shunts.

Key points to remember: - Ischaemic: $pO_2 < 30$, $pCO_2 > 60$, $pH < 7.25$ – emergency. - Phenylephrine 100–500 $\mu\text{g/mL}$, 1 mL q3–5 min up to 1 h. - Distal shunts (Winter, Ebbehøj, T, Al-Ghorab) before proximal (Quackels, Grayhack). - Prosthesis for >36 h; high-flow → observe/embolise.

II-3b. Renal tubular acidosis

Type: SAQ (10 marks) | ~550 words

Definition

- Group of disorders with **hyperchloraemic, normal anion-gap metabolic acidosis** due to impaired renal bicarbonate reabsorption or H⁺ excretion, with relatively preserved GFR.

Comparison of types

Feature	Type 1 (distal)	Type 2 (proximal)	Type 4 (hyperkalaemic)
Defect	Impaired H ⁺ secretion in collecting duct (H ⁺ -ATPase/H ⁺ -K ⁺ -ATPase/anion exchanger; gradient defect)	Impaired HCO ₃ ⁻ reabsorption in PCT (↓ threshold)	Aldosterone deficiency/resistance → ↓ NH ₄ ⁺ excretion
Urine pH	>5.5 always	<5.5 once serum HCO ₃ falls below threshold; >5.5 during bicarbonate loading	<5.5
Serum K	Low	Low	High
Serum HCO ₃	Very low (<10-15 possible)	12-20 mmol/L	>15 mmol/L
Urine citrate	Low	Normal/high	Normal
Hypercalciuria	Yes	Yes (mild)	No
Stones / nephrocalcinosis	Common (calcium phosphate)	Rare	Absent
Associated	Osteomalacia, rickets	Fanconi syndrome, rickets	-
Causes	Genetic; Sjogren's, SLE, RA; amphotericin, lithium, ifosfamide; hypercalciuria, MSK, obstructive uropathy	Fanconi, cystinosis, Wilson, multiple myeloma, heavy metals, acetazolamide, tenofovir, ifosfamide	Diabetic nephropathy, obstructive uropathy , Addison's, ACEi/ARB, spironolactone, trimethoprim, NSAIDs, heparin, calcineurin inhibitors
Treatment	Potassium citrate/alkali 1-2 mEq/kg/d	Large-dose alkali (10-15 mEq/kg/d) + K; thiazide	Low K diet, loop diuretic, fludrocortisone , stop offending drugs, bicarbonate

- Type 3:** rare (carbonic anhydrase II deficiency) – mixed proximal + distal, with osteopetrosis and cerebral calcification.
- Incomplete dRTA:** normal serum HCO₃ but inability to acidify urine – seen in many recurrent calcium phosphate stone formers.

Diagnosis

- ABG/serum electrolytes: normal AG metabolic acidosis ($AG = Na - [Cl + HCO_3]$).
- Urine anion gap (UAG) = (Na + K) - Cl:** surrogate for NH₄⁺ excretion.
 - Negative** → appropriate NH₄⁺ excretion (GI bicarbonate loss, proximal RTA).
 - Positive** → impaired NH₄⁺ excretion (distal RTA, type 4).
- Fresh **urine pH** (pH meter, morning sample under oil).

- **Ammonium chloride loading test** (0.1 g/kg orally; urine pH hourly ~6 h): failure to lower urine pH **<5.5** = distal RTA.
- **Furosemide-fludrocortisone test** (furosemide 40 mg + fludrocortisone 1 mg): better tolerated alternative; failure to acidify <5.5 confirms dRTA.
- Fractional excretion of HCO_3^- >15% during bicarbonate loading = proximal RTA.
- Urine-blood pCO_2 gradient during bicarbonate loading (<20 mmHg in dRTA).
- Imaging: US/CT for nephrocalcinosis; 24-h urine citrate/calcium.

Urological relevance

- dRTA → **recurrent calcium phosphate stones, nephrocalcinosis**, hypocitraturia; also predisposes to infection stones.
- **Medullary sponge kidney** often associated with incomplete dRTA.
- **Obstructive uropathy** (PUV, BPO) → type 4/hyperkalaemic distal RTA – improves after relief of obstruction.
- **Topiramate/acetazolamide** (carbonic anhydrase inhibitors) → CaP stones.
- **Urinary diversion** (ureterosigmoidostomy, ileal conduit, neobladder) → hyperchloraemic acidosis mimicking RTA.
- Post-transplant RTA (calcineurin inhibitors).
- Potassium citrate preferred over sodium bicarbonate (sodium increases calcium excretion).

Recent advances

- Genetic testing (SLC4A1, ATP6V1B1, ATP6V0A4) for inherited dRTA; prolonged-release alkali formulations (e.g. ADV7103) for dRTA.

Diagram to draw: Nephron diagram showing site of defect – PCT (type 2), collecting duct α -intercalated cell (type 1), aldosterone action (type 4).

Key points to remember: - Type 1: urine pH >5.5, hypokalaemia, hypocitraturia, CaP stones/nephrocalcinosis. - Type 4: hyperkalaemia, no stones; seen in obstructive uropathy. - Positive UAG = distal RTA; NH_4Cl 0.1 g/kg – pH fails to fall <5.5. - Treat dRTA with potassium citrate.

II-3c. Millin's prostatectomy

Type: SAQ (10 marks) | ~550 words

Definition / History

- **Retropubic extravesical simple prostatectomy:** enucleation of the obstructing adenoma through a **transverse incision in the anterior prostatic capsule**, without opening the bladder.
- Described by **Terence Millin (Lancet, 1945)**.
- Contrast: **Freyer's transvesical (suprapubic) prostatectomy** – adenoma enucleated through an opened bladder.
- Campbell: Simple prostatectomy (open and robotic) chapter.

Indications

- Large gland, generally **>80-100 mL** (AUA), when endoscopic enucleation (HoLEP) unavailable.
- Indications for BPH surgery (retention, recurrent UTI/haematuria, renal impairment, bladder stones, refractory LUTS).
- Concomitant pathology: inguinal hernia repair through same incision.
- **Best for:** large lateral lobes with small/no median lobe.
- **Relative contraindications:** large intravesical median lobe, bladder diverticulum/large stones (transvesical preferred), small fibrous gland, previous pelvic surgery/radiation, prostate cancer.

Steps

1. Supine/slight Trendelenburg; urethral catheter; Pfannenstiel or lower midline incision.
2. Extraperitoneal approach – develop **space of Retzius**; clear periprostatic fat.
3. **Haemostatic control:** suture ligatures across the anterior capsule proximal and distal to the planned incision (controls dorsal venous plexus of Santorini) ± ligation of lateral pedicles at bladder neck.
4. **Transverse capsulotomy** ~1-2 cm distal to bladder neck with diathermy.
5. Develop plane between adenoma and surgical capsule with scissors, then **finger enucleation**.
6. **Divide urethra at apex under direct vision** (protects external sphincter).
7. Inspect bladder neck and ureteric orifices; **wedge excision of posterior bladder neck lip** (prevents bladder neck contracture); advance mucosa (trigonisation) as needed.
8. Haemostasis: figure-of-eight sutures at **5 and 7 o'clock** at bladder neck.
9. Place 20-22 Fr three-way catheter; **close capsule** watertight; drain in Retzius; closure.

Advantages over Freyer (transvesical)

Aspect	Millin (retropubic)	Freyer (transvesical)
Bladder	Not opened	Opened
Apex/urethral division	Under direct vision – better continence	Blind finger fracture at apex
Haemostasis	Prostatic fossa directly visible	Fossa less visible; packing/purse-string needed
Bladder pathology (stone, diverticulum)	Difficult	Easily addressed
Large median lobe	Difficult	Ideal

Aspect	Millin (retropubic)	Freyer (transvesical)
Catheter time/stay	Shorter	Longer (suprapubic + urethral catheter)
Specific risk	Santorini bleeding, osteitis pubis	Bladder spasm, urinary leak/SP fistula

Complications

- Intra/early: **haemorrhage/transfusion**, clot retention, rectal injury (rare), UTI/sepsis, urinary leak, DVT/PE, wound infection.
- Late: **retrograde ejaculation** (most), bladder neck contracture, urethral stricture, stress incontinence (uncommon), ED (uncommon), **osteitis pubis**, residual adenoma.
- Excellent durable outcomes – large IPSS and Qmax improvement, low retreatment.

Modern alternatives

- **HoLEP/ThuLEP/bipolar enucleation** – endoscopic equivalent, size-independent, less bleeding and shorter stay; EAU recommends as first choice for >80 mL where available.
- **Robotic-assisted simple prostatectomy (RASP)** – transvesical or Millin-type extravesical approaches; lower blood loss and shorter stay than open, similar functional outcome; allows concomitant diverticulectomy/stone removal.
- Laparoscopic simple prostatectomy; Aquablation (80–150 mL).

Diagram to draw: Sagittal section showing transverse capsulotomy below bladder neck with finger enucleation plane between adenoma and surgical capsule.

Key points to remember: - Millin 1945: retropubic, transverse anterior capsulotomy, bladder not opened. - Apical urethral division under vision; 5 and 7 o'clock haemostatic sutures; bladder neck wedge. - Freyer (transvesical) preferred for median lobe, stones, diverticulum. - HoLEP and robotic simple prostatectomy are modern alternatives for >80–100 mL.

II-3d. Energy sources in laparoscopic urology

Type: SAQ (10 marks) | ~550 words

Introduction

- Energy devices achieve **cutting, coagulation and vessel sealing**; choice affects blood loss, speed and thermal injury.
- Tissue effects: ~60 °C – protein denaturation (coagulation); ~100 °C – water vaporization (cutting); >200 °C – carbonization.
- Campbell: Basics of laparoscopic and robotic surgery chapter.

Types of energy sources

Device	Principle	Uses / features	Limitations
Monopolar electrosurgery	RF current (>100 kHz, typically 300 kHz–3 MHz) passes from active electrode through patient to return pad; cut (continuous, low-voltage) vs coag (interrupted, high-voltage) waveforms	Cheap, versatile; hook, scissors	Greatest lateral spread; stray current injury; pacemaker interference
Conventional bipolar	Current between two jaws only	Safer near nerves/bowel (e.g. nerve-sparing, small vessels)	Slower; no cutting; sticking/charring
Advanced bipolar vessel sealers (LigaSure, ENSEAL)	Pressure + impedance-controlled bipolar energy fusing collagen/elastin; integrated blade	Seal vessels up to 7 mm ; nephrectomy, adrenalectomy, RPLND, cystectomy pedicles	Cost; lateral spread ~1–3 mm; jaw heat
Ultrasonic (Harmonic scalpel)	Blade vibrates at ~55.5 kHz → mechanical energy, protein denaturation (coaptive coagulation) at lower temperatures (<100 °C)	Simultaneous cut + seal; vessels up to 5 mm (newer models up to 7 mm); no current through patient; less smoke (mist)	Active blade remains hot; slower on thick tissue
Hybrid (Thunderbeat)	Ultrasonic + advanced bipolar	Fast dissection + seals up to 7 mm	Cost
Lasers	Holmium:YAG (2,120 nm), thulium (2,013 nm/TFL 1,940 nm), KTP (532 nm), diode	Partial nephrectomy (experimental), stricture incision, laser-assisted pyeloplasty, endourology	Smoke, eye safety
Argon beam coagulator	Monopolar current conducted through ionized argon gas – non-contact, superficial uniform eschar	Haemostasis of renal/liver/splenic parenchymal bed (partial nephrectomy)	Gas embolism (vent pneumoperitoneum), pressure rise
Hydro-jet (water-jet dissection)	High-pressure water jet separates parenchyma, preserves vessels	Partial nephrectomy, nerve-sparing	Not haemostatic
Others	Radiofrequency ablation, microwave, cryoablation	Ablation of small renal masses	–

Safety - mechanisms of electrosurgical injury

- Many injuries unrecognised intra-operatively (occur outside the field of view) → delayed bowel perforation presenting days later. 1. **Insulation failure** – defects in instrument

insulation (repeated sterilisation, high-voltage coag) → current leaks to bowel. 2. **Direct coupling** – active electrode touches another metal instrument/laparoscope in contact with tissue. 3. **Capacitive coupling** – current induced in adjacent conductor through intact insulation; worse with high-voltage coag, long activation, small-port with metal; **avoid hybrid trocars** (metal cannula with plastic anchor) – use all-metal or all-plastic systems. 4. **Direct thermal spread** – lateral spread to ureter, nerves (neurovascular bundles), bowel; residual heat of instrument tips (ultrasonic blade). 5. **Return electrode burns** – poor pad contact (prevent by **return electrode contact quality monitoring, REM**). 6. **Smoke plume**; fires in oxygen-rich environment.

Safety measures

- **Active Electrode Monitoring (AEM)** – shields instrument and shuts generator if insulation failure or capacitive coupling occurs.
- Use lowest effective power, cut waveform where possible, brief activation, avoid open-circuit activation.
- Keep active tip in view; do not activate near/in contact with other metal instruments.
- Inspect insulation; bipolar near critical structures; avoid thermal energy near neurovascular bundle (athermal nerve-sparing in prostatectomy).
- Pacemaker/ICD: bipolar preferred, cardiology input.

Recent advances

- Smart generators with real-time impedance feedback; robotic vessel sealers (e.g. SynchroSeal, Vessel Sealer Extend); ultrasonic devices sealing up to 7 mm.

Diagram to draw: Monopolar circuit vs bipolar circuit; capacitive coupling across a hybrid trocar.

Key points to remember: - LigaSure/advanced bipolar: vessels up to 7 mm; Harmonic: up to 5 mm at ~55.5 kHz. - Stray current injury: insulation failure, direct coupling, capacitive coupling. - Avoid hybrid trocars; use AEM and REM pads. - Argon beam – risk of gas embolism.

II-3e. Role of vascular grafts in urology

Type: SAQ (10 marks) | ~550 words

Introduction

- Vascular grafts replace, bypass or extend vessels when primary repair is impossible; increasingly needed in transplantation and extended oncological resections.
- Campbell: Renal transplantation, Renovascular hypertension, and Renal tumours (IVC thrombus) chapters.

Types of grafts

Type	Examples	Advantages	Disadvantages
Autologous vein	Great saphenous vein, gonadal vein, donor/recipient iliac or renal vein	Low infection, best long-term patency in small vessels	Limited length/calibre, aneurysmal dilatation
Autologous artery	Internal iliac (hypogastric) artery, inferior epigastric artery, donor aortic (Carrel) patch	Ideal for renal artery reconstruction, matching size; growth in children	Limited availability
Synthetic	PTFE (e-PTFE, Gore-Tex), Dacron (polyester)	Off-the-shelf, sizes available, large vessels (IVC, aorta)	Infection, thrombosis in low-flow/small calibre, contraindicated in contaminated field
Biological / cadaveric	Cryopreserved cadaveric vessels, deceased-donor iliac artery/vein; bovine pericardium patch	Useful in infected fields, deceased-donor transplant extension	Availability, degeneration
Patches	Bovine pericardium, PTFE patch	Partial IVC wall defects (cavotomy closure)	-

Applications in urology

1. Renal transplantation

- **Multiple donor arteries** (living donor): bench reconstruction – side-to-side conjoining, end-to-side implantation of small polar artery onto main artery, or inferior epigastric anastomosis to a lower-pole artery.
- Short renal vein (right kidney): extension with donor gonadal vein, saphenous vein or deceased-donor iliac/IVC segment.
- Deceased donor: Carrel aortic patch, donor iliac Y-grafts for reconstruction (e.g. pancreas/multiple vessels).
- Recipient atherosclerotic/aneurysmal iliacs: synthetic aorto-iliac bypass prior to/with transplant.

2. Renovascular surgery

- Renal artery stenosis (fibromuscular dysplasia, atherosclerosis), **renal artery aneurysm**, AV malformation: aortorenal bypass with saphenous vein, hypogastric artery or PTFE; splenorenal/hepatorenal bypass; ex vivo bench repair.

3. Renal autotransplantation

- Complex ureteric loss, renal artery branch disease, loin-pain haematuria; vascular extension grafts as needed.

4. RCC with IVC tumour thrombus (Neves-Zincke levels)

- Most managed by cavotomy and primary closure; **patch cavoplasty** (bovine pericardium/PTFE) if lumen narrowed >50%.
- **IVC resection and replacement** with ring-reinforced **PTFE graft** when wall invaded and collaterals inadequate; infrarenal IVC can often be ligated without graft when collaterals developed.
- Level III-IV thrombi may need cardiopulmonary bypass.

5. Retroperitoneal tumours / post-chemotherapy RPLND

- Residual masses encasing the aorta (testicular GCT) – **aortic resection with Dacron/PTFE interposition graft**; IVC resection ± reconstruction; renal artery reimplantation.
- Retroperitoneal sarcomas involving iliac vessels/IVC.

6. Vascular access

- **AV graft (PTFE)** for haemodialysis when native fistula impossible.

7. Penile revascularisation

- Inferior epigastric artery to dorsal artery/deep dorsal vein (microsurgical) – young men with arteriogenic ED after pelvic trauma; rarely performed.

8. Vascular injury

- Iatrogenic injury during laparoscopic nephrectomy, pelvic lymphadenectomy, cystectomy; trauma – interposition saphenous/PTFE graft.

Principles

- Prefer **autologous** in contaminated/urine-exposed fields and small vessels; synthetic for large-calibre high-flow vessels.
- Tension-free, appropriately sized anastomosis; intimal-to-intimal apposition; spatulation; fine non-absorbable monofilament (polypropylene).
- Systemic heparinisation during clamping; postoperative antiplatelet/anticoagulation as indicated.
- Avoid kinking; omental wrap to isolate synthetic graft from urinary tract/bowel.
- Perioperative antibiotics; Doppler surveillance.

Complications

- Thrombosis/occlusion, anastomotic stenosis, haemorrhage, pseudoaneurysm, **graft infection** (difficult – often needs explantation), graft-enteric/urinary fistula, lymphocele, distal embolism, limb oedema (IVC ligation).

Recent advances

- Robotic IVC thrombectomy and robotic bench/vascular reconstruction; bioengineered acellular vessels (human acellular vessel) for dialysis access/trauma.

Diagram to draw: Bench reconstruction of two renal arteries (side-to-side conjoining and end-to-side implantation) before transplant.

Key points to remember: - Autologous (saphenous, hypogastric, gonadal) for small vessels/contaminated fields; PTFE/Dacron for IVC/aorta. - Transplant: Carrel patch, bench reconstruction of multiple arteries, gonadal/IVC extension of short right vein. - IVC: patch cavoplasty if narrowed; PTFE replacement if wall invaded. - Post-chemo RPLND may need aortic graft replacement.

II-3f. Transrectal ultrasonography

Type: SAQ (10 marks) | ~550 words

Introduction / Physics

- Real-time imaging of prostate, seminal vesicles and adjacent structures through the rectal wall.
- Piezoelectric crystals convert electrical to sound energy and back; higher frequency = better resolution but less penetration.
- Probes: **6-10 MHz (commonly 7.5 MHz), biplanar** (transverse + sagittal) – end-fire or side-fire; colour/power Doppler, elastography.
- Campbell: Prostate biopsy – techniques and imaging chapter.

Technique

- Left lateral decubitus (or lithotomy); DRE first; enema/rectal preparation.
- Systematic scanning: transverse base→apex, then sagittal midline → laterally.

Normal anatomy (McNeal zones)

Zone	% glandular volume	Echogenicity	Clinical
Peripheral zone	~70%	Homogeneous, isoechoic	~70% of cancers
Transition zone	~5-10% (enlarges in BPH)	Heterogeneous, hypoechoic	BPH, ~20-25% cancers
Central zone	~25%	Slightly hyperechoic	~5-10% cancers
Anterior fibromuscular stroma	Non-glandular	Hypoechoic	-

- Seminal vesicles: paired "bow-tie" hypoechoic structures; ejaculatory ducts; verumontanum.

Measurements

- **Prostate volume (ellipsoid formula) = $0.52 \times L \times W \times H$** (1 mL $\approx$ 1 g).
- **PSA density (PSAD) = PSA/volume**; >0.15 ng/mL/mL suspicious; used in deciding biopsy after MRI (e.g. PI-RADS 3).
- Classic cancer appearance: **hypoechoic peripheral-zone lesion** (low specificity; many cancers isoechoic).

Indications

- **Guidance for prostate biopsy** (raised PSA, abnormal DRE, active surveillance, MRI lesion).
- Volume estimation – planning BPH surgery and choice of modality, PSAD.
- **Ejaculatory duct obstruction**: dilated seminal vesicles (>1.5 cm AP), dilated ejaculatory ducts, midline (utricular/Müllerian) cysts, calcification; TRUS-guided seminal vesicle aspiration/vesiculography.
- Prostatic abscess – diagnosis and transrectal/transperineal drainage.
- **Brachytherapy** (volume study, seed placement – transperineal under TRUS), **HIFU**, **cryotherapy**, focal therapy, fiducial marker/rectal spacer placement.
- Staging (ECE/SVI) – largely replaced by MRI.

TRUS-guided biopsy

- **Systematic 12-core** (sextant medial + lateral at base, mid, apex) – lateral (far-lateral PZ) cores improve detection; $\pm$ targeted cores.
- Anaesthesia: **periprostatic nerve block** (lidocaine at junction of prostate base and seminal vesicles).
- Stop anticoagulants per risk (aspirin often continued).

Complications

- Haematospermia (most common), haematuria, rectal bleeding.
- **Infection** – UTI, prostatitis, **sepsis** (rising due to fluoroquinolone-resistant E. coli); urinary retention, vasovagal reaction.

Prophylaxis (transrectal route)

- Rectal cleansing with **povidone-iodine**; targeted prophylaxis based on rectal swab culture or augmented prophylaxis; fluoroquinolones restricted (EMA 2019) – EAU no longer recommends them as default.

Current shift

- **EAU (2024/25): transperineal biopsy recommended** over transrectal because of significantly lower infection risk; antibiotic prophylaxis may be omitted for transperineal biopsy. Freehand LA transperineal access systems (PrecisionPoint, CamPROBE) enable office-based procedure.
- **Pre-biopsy mpMRI and MRI-targeted biopsy** (cognitive, software fusion, in-bore):
 - **PROMIS (2017)**: mpMRI sensitivity for csPCa 93% vs TRUS biopsy 48%.
 - **PRECISION (2018)**: MRI-targeted biopsy detected more csPCa (38% vs 26%) and less insignificant cancer than standard TRUS biopsy.

Recent advances

- **Micro-ultrasound (29 MHz, ExactVu; PRI-MUS scoring)** – higher resolution; **OPTIMUM RCT (2025)** reported micro-US-guided biopsy non-inferior to MRI-targeted biopsy for csPCa detection.
- Multiparametric US (contrast-enhanced, shear-wave elastography), AI-assisted TRUS segmentation, fusion platforms.

Diagram to draw: Transverse and sagittal TRUS views of zonal anatomy; 12-core biopsy template.

Key points to remember: - Biplanar probe 6–10 MHz; volume = $0.52 \times L \times W \times H$; PSAD >0.15 suspicious. - 12-core systematic biopsy with periprostatic block. - EAU: transperineal preferred (lower sepsis); FQ prophylaxis restricted. - PROMIS and PRECISION established MRI-first pathway.